

Project 1 Data Card

Apple Financial Dataset

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Card ID

Primary Dataset	<i>Apple Financial Dataset (1980-2026)</i>
Description	<i>All-in-One Apple Stock Dataset: Prices, Financials, and ML Features (1980-2026)</i>
Supplemental Dataset 1	<i>Apple Stock (AAPL): Historical Financial News Data</i>
Description	<i>A comprehensive dataset of financial news on Apple Inc. (AAPL) from 2016 to 2024</i>
Supplemental Dataset 2	<i>Apple iPhone Product & Revenue Dataset (2007–2025)</i>
Description	<i>A yearly dataset capturing iPhone product releases, estimated sales, pricing, and revenue contribution from 2007 to 2025</i>
Supplemental Dataset 3	<i>Global iPhone & Smartphone Market Dataset (2011–2023)</i>
Description	<i>A global smartphone market dataset covering iPhone sales, users, and iOS vs. Android market share trends from 2011 to 2023</i>

Add more rows to the above table for each additional supplemental dataset.

Datasets at a glance

Primary dataset

The primary dataset contains 11,879 examples representing daily historical Apple stock records in the period from 12/12/1980 to 01/27/2026.

Fields (if applicable)

Field name	Data Type	Description
date	Date	Trading day timestamp; ranges 12/12/1980–01/27/2026. Example: 2024-11-22
close_price	Float	Closing stock price for the quarter; primary indicator of Apple's market value over time. Example: 0.116784.
open_price	Float	Opening stock price for the quarter; used to measure price movement relative to the closing price. Example: 0.098389.
total_volume	Integer	Number of shares traded; ranges 1M → 200M+ depending on year. Example: 42,300,000
avg_daily_return	Float	Average percentage change in price from one trading day to the next within the quarter; measures growth and momentum. Example: 0.015527
return_volatility	Float	Standard deviation of daily returns within the quarter; quantifies risk and the intensity of price fluctuations. Example: 0.049939

Supplemental dataset 1

The supplemental dataset contains news articles related to Apple Inc. (AAPL), representing financial and market-related coverage during the period from 01/01/2016 to 12/31/2024.

Fields (if applicable)

Field name	Data Type	Description
Date	Date	Publication date of the news article Example: 2024-11-26T00:00:00+00:00
Headline	String	Title of the news article Example: Could Investing \$1,000 in Apple Make You a Millionaire?
Source	String	News publisher or media outlet Example: https://www.fool.com/investing/2024/11/26/could-investing-1000-apple-make-you-millionaire/

Summary	String	Short summary or excerpt of the article Example: A Complete Guide to College Savings Accounts in 2024
Sentiment	Float	Sentiment score associated with the article Example: 0

Supplemental dataset 2

The supplemental dataset contains annual observations representing the iPhone's product releases, sales performance, and revenue impact on Apple during the period from 01/01/2007 to 12/31/2025.

Fields (if applicable)

Field name	Data Type	Description
Year	Integer	Calendar year of observation Example: 2007
iPhone_Model_Released	String	Flagship iPhone model(s) released in the given year Example: iPhone 3G
Units_Sold_Millions	Float	Estimated number of iPhones sold in the year (in millions) Example: 1.99
Average_Selling_Price_USD	Float	Estimated average selling price of iPhones (USD) Example: 499
Annual_iPhone_Revenue_Billions	Float	Estimated total iPhone revenue for the year (USD billions) Example: 0.95

Supplemental dataset 3

The supplemental dataset contains annual observations representing global iPhone sales, user adoption, and smartphone market share trends during the period from 01/01/2011 to 12/31/2023.

Fields (if applicable)

Field name	Data Type	Description
Year	Integer	Calendar year of observation Example: 2023

Global_iPhone_Users_Millions	Float	Estimated number of iPhone users worldwide (in millions) Example: 1460000000
iPhone_Sales_Millions	Float	Annual global iPhone unit sales Example: 234600000
iOS_Market_Share_Percent	Float	Global smartphone market share of iOS (%) Example: 58.10
Android_Market_Share_Percent	Float	Global smartphone market share of Android (%) Example: 41.46

Additional notes

Provide any additional notes on the data here (e.g. changes in data collection and labeling practices).

Usage information

Use(s)

Primary Data

Usage for the dataset includes time-series forecasting, or analyzing the correlation between financial reports and stock price movements and for educational and research purposes. Discouraged use cases are using data for real-time trading and investment decisions, official financial statements, and high-stakes predictive modeling.

1st Supplementary Data

Usage for the dataset includes analyzing financial news sentiment related to Apple Inc., studying the relationship between media coverage and stock price movements, and developing NLP-based sentiment analysis models for educational and research purposes. Discouraged use cases include real-time trading systems, automated investment decision-making, and high-stakes financial prediction.

2nd Supplementary Data

Usage for the dataset includes analyzing long-term iPhone sales and revenue trends, evaluating Apple's product and pricing strategy, and supporting educational, exploratory, and business intelligence analyses. Discouraged use cases include real-time sales forecasting, official financial reporting, and investment decision-making.

3rd Supplementary Data

Usage for the dataset includes analyzing global smartphone market trends, comparing iOS and Android market share, studying user adoption patterns, and conducting market competition research. Discouraged

use cases include real-time market prediction, commercial market intelligence products, and high-stakes financial or strategic decision-making.

Licensing

Primary Data

CC BY-NC-SA 4.0

Data can be shared and adapted

Credit must be given to the author, only permitted for non-commercial use, modified dataset falls under the same license.

1st Supplementary Data

CC BY-NC-SA 4.0

The data may be shared and adapted for non-commercial purposes with proper attribution to the dataset author. Any modified or derived datasets must be distributed under the same license, and changes should be clearly documented.

2nd Supplementary Data

CC BY-NC-SA 4.0

The data may be shared and adapted for non-commercial purposes with proper attribution to the original author. Any modified versions must be distributed under the same license, and changes should be clearly indicated.

3rd Supplementary Data

CC BY-NC-SA 4.0

The data may be shared and transformed for non-commercial use with appropriate credit. Derivative works must carry the same license, and usage for commercial or proprietary applications is restricted.

Privacy considerations

Primary Data

Dataset only includes publicly available data since Apple is a publicly traded company which means all of their financials are publicly accessible. No PII is included.

1st Supplementary Data

The dataset consists of publicly available financial news articles and metadata related to Apple Inc. No personally identifiable information (PII) is included. All content is derived from public news sources and analyzed at the article level without reference to individual users.

2nd Supplementary Data

The dataset contains aggregated, product-level and company-level information about iPhone releases, estimated sales, and revenue. No personally identifiable information (PII) is included, and all figures are reported or estimated at an annual level.

3rd Supplementary Data

The dataset includes aggregated global statistics on smartphone users, sales, and market share. No individual-level data or personally identifiable information (PII) is present, and all values are reported in summarized form.

Format

The primary dataset is provided in a CSV file.

The first supplemental dataset is provided in a CSV file.

The second supplemental dataset is provided in a xlsx file.

The third supplemental dataset is provided in a CSV file.

Source and collection process

Origin

Primary Data

Yahoo Finance- <https://finance.yahoo.com/quote/AAPL/history/>

Nasdaq Apple Official Historical Data Table - <https://www.nasdaq.com/market-activity/stocks/aapl/historical>

- 1st Supplementary Data

Apple Stock (AAPL): Historical Financial News Data

Source: Kaggle

<https://www.kaggle.com/datasets/Apple AAPL financial news dataset, 2016–2024>

- 2nd Supplementary Data

Apple iPhone Product & Revenue Dataset (2007–2025)

Source: Kaggle

<https://www.kaggle.com/datasets/atharvamohite4631/apple-data-set-salesrevenuestock-priceprice>

- 3rd Supplementary Data

Global iPhone & Smartphone Market Dataset (2011–2023)

Source: Kaggle

<https://www.kaggle.com/datasets/mohamedfahim003/global-iphone-and-smartphone-market-2011-2023>

Original datasets author/publisher

Primary Data

Yahoo Finance - Yahoo Inc. (Yahoo Finance Data Team)

Nasdaq Apple Official Historical Data Table - Nasdaq

- 1st Supplementary Data

Kaggle Community Dataset (published via Kaggle)

- 2nd Supplementary Data

Atharva Mohite (Kaggle dataset publisher)

- 3rd Supplementary Data

Mohamed Fahim (Kaggle dataset publisher)

Original datasets point of contact

Primary Data

Yahoo Finance

- Yahoo Finance Support Portal: <https://help.yahoo.com>
- General Yahoo Customer Care: <https://help.yahoo.com/contact>

Nasdaq

- Nasdaq Market Data Support: <https://www.nasdaq.com/about/contact-us>
- Market Data Inquiries Email: datasales@nasdaq.com

- 1st Supplementary Data

Dataset: Apple Stock (AAPL) Historical Financial News Data

Publisher: Frank Ossai (Kaggle)

- 2nd Supplementary Data

Dataset: Apple Data Set (Sales, Revenue, Stock Price)

Publisher: Atharva Mohite (Kaggle)

- 3rd Supplementary Data

Dataset: Global iPhone & Smartphone Market (2011–2023)

Publisher: Mohamed Fahim (Kaggle)

Original metadata

Primary Data

Apple Comprehensive Financial Dataset (1980–2026) — Metadata Page

<https://www.kaggle.com/datasets/adamvakar/apple-comprehensive-financial-dataset-1980-2026>

- 1st Supplementary Data

Apple Stock (AAPL) Historical Financial News Data

Metadata Link:

<https://www.kaggle.com/datasets/frankossai/apple-stock-aapl-historical-financial-news-data>

- 2nd Supplementary Data

Apple Data Set (Sales, Revenue, Stock Price)

Metadata Link:

<https://www.kaggle.com/datasets/atharvamohite4631/apple-data-set-salesrevenuestock-priceprice>

- 3rd Supplementary Data

Global iPhone & Smartphone Market (2011–2023)

Metadata Link:

<https://www.kaggle.com/datasets/mohamedfahim003/global-iphone-and-smartphone-market-2011-2023>

Collection process

- Primary Data

Yahoo

- Yahoo Finance - Price Data Adjustments & Methodology - <https://help.yahoo.com/kb/SLN28256.html>
- Yahoo Finance Historical Data (AAPL) - Source Table - <https://finance.yahoo.com/quote/AAPL/history/>

Nasdaq - Official Market Data Methodology

- Nasdaq Market Data Products - Collection & Processing Overview - <https://www.nasdaq.com/solutions/market-data>
- Nasdaq Data Collection & Dissemination - Technical Documentation - <https://www.nasdaqtrader.com/Trader.aspx?id=MarketData>
- Nasdaq AAPL Historical Data (Source Table) - <https://www.nasdaq.com/market-activity/stocks/aapl/historical>

- 1st Supplementary Data

https://www.kaggle.com/datasets/frankossai/apple-stock-aapl-historical-financial-news-data?utm_source=chatgpt.com

Collected and compiled by the dataset author using publicly available financial news articles related to Apple Inc., as described on the Kaggle dataset page.

- 2nd Supplementary Data

<https://www.kaggle.com/datasets/atharvamohite4631/apple-data-set-salesrevenuestock-priceprice?resource=download>

Collected and synthesized from publicly available Apple financial reports and market data, with post-2018 figures estimated by the author based on revenue and pricing trends, as documented on the Kaggle dataset page.

- 3rd Supplementary Data

https://www.kaggle.com/datasets/mohamedfahim003/global-iphone-and-smartphone-market-2011-2023?utm_source=chatgpt.com

Aggregated by the dataset author from publicly available global smartphone market reports and industry statistics, as described on the Kaggle dataset page.

Online link(s) to information on data source's collection and labeling processes.

Collection or access date

- Primary Data

Accessed: January 2026

- 1st Supplementary Data

Accessed: January 2026

- 2nd Supplementary Data

Accessed: January 2026

- 3rd Supplementary Data

Accessed: January 2026

Refresh date/schedule

- Primary Data

Last refresh Jan 3, 2026. Scheduled to refresh annually.

- 1st Supplementary Data

Not specified; dataset appears to be a fixed historical snapshot.

- 2nd Supplementary Data

Not specified; single published version with projected values.

- 3rd Supplementary Data

Not specified; dataset represents a fixed historical range.

Last refresh date and refresh cadence, or N/A if there is only one version.

Exploration Study

Questions driving the exploration study

- 1) Is daily news sentiment associated with Apple's daily stock return?
- 2) Do extreme sentiment days correspond to higher trading volume?
- 3) Do sentiment spikes precede larger price volatility?
- 4) How has Apple's stock price evolved over time?
- 5) Is iPhone revenue dependency associated with stock performance?
- 6) Does relative iOS market share (vs Android) relate to Apple's stock performance?

Descriptive statistics

Primary Dataset:

Variable	Count	Mean	Std	IQR	Min	Max
close_price	182	29.78	61.57	21.88	0.043634	271.859985
open_price	182	28.39	59.28	21.23	0.043634	272.260010
trading_volume	182	1.93e10	1.62e10	1.72e10	7.33e8	8.52e10
avg_daily_return	182	0.001106	0.003928	0.005085	-0.010246	0.015527
return_volatility	182	0.025439	0.010922	0.014315	0.009645	0.077053

Supplemental Dataset 1

Variable	Count	Mean	Std	IQR	Min	Max
sentiment_score	29,737	0.50	0.58	0.99	-1.00	1.00

Supplemental Dataset 2

Variable	Count	Mean	Std	IQR	Min	Max
units_sold_in_year_millions	19	157.36	84.39	123.70	1.90	231.20
average_selling_price	19	719.95	157.17	225	450	945
annual_iphone_revenue_billions	19	124.63	75.99	120.58	0.95	218.30

Supplemental Dataset 3

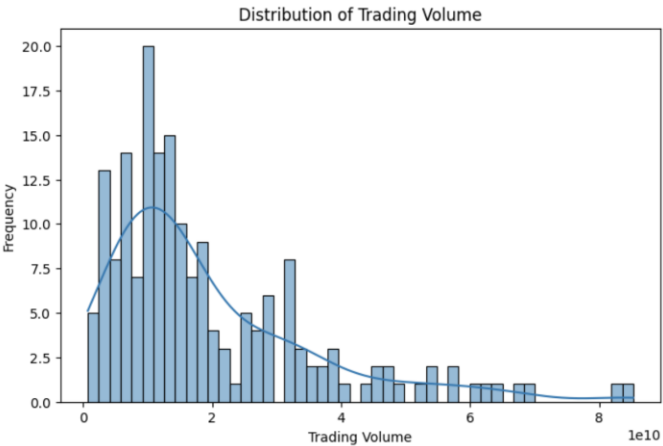
Variable	Count	Mean	Std	IQR	Min	Max
no_of_iphone_users	13	7.75e08	4.31e08	5.58e08	1.15e08	1.46e09
no_of_iphone_sold	13	1.95e08	4.31e07	3.50e07	9.31e07	2.36e08
ios_market_share	11	55.09	2.83	4.43	50.85	59.54
android_market_share	11	43.06	2.34	3.67	39.25	46.42

Distribution plots



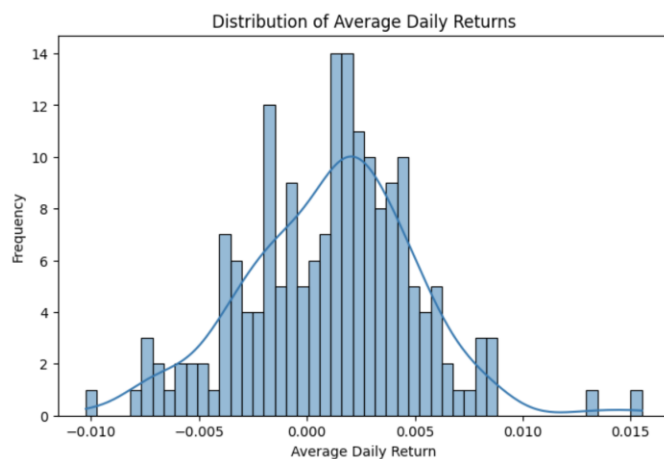
closing_price

The distribution of Apple's closing stock price is highly right-skewed, indicating that stock prices were relatively low during earlier years and increased significantly over time. This pattern reflects Apple's long-term financial growth and increasing market valuation.



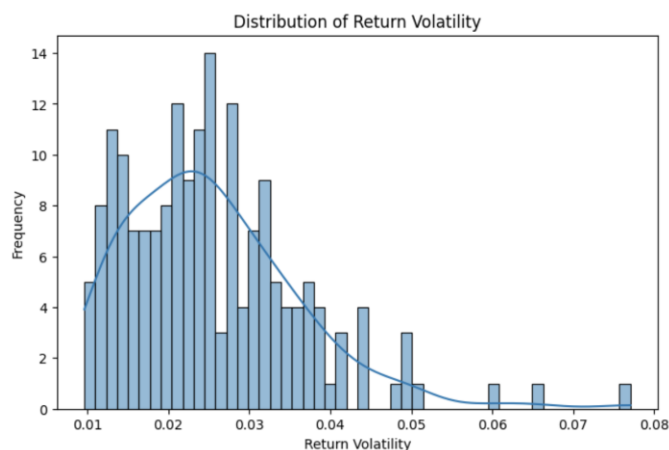
total_volume

Trading volume shows high variability with several extreme spikes. This suggests periods of increased investor activity, which may correspond to major company announcements, product launches, or broader market events. The data is skewed to the right, suggesting an overall lower trading volume apart from a few outliers.



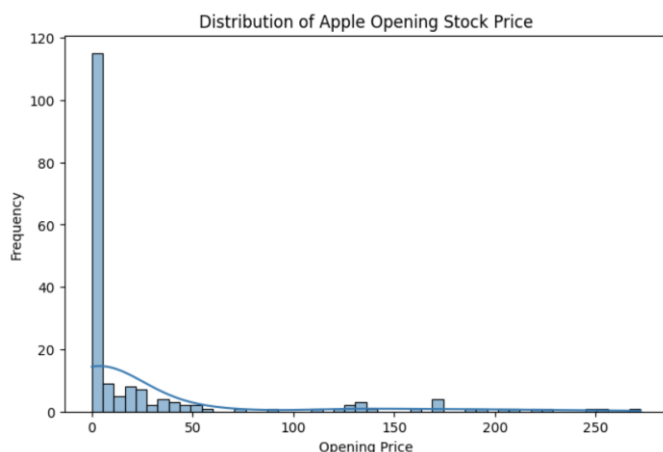
avg_daily_return

The distribution of average daily returns is centered around zero, showing that gains and losses occur with similar frequency. Its relatively symmetrical bell shape reflects typical stock-market return behavior. Most values cluster close to zero, indicating low day-to-day volatility. The light tails show that extreme return swings are rare.



return_volatility

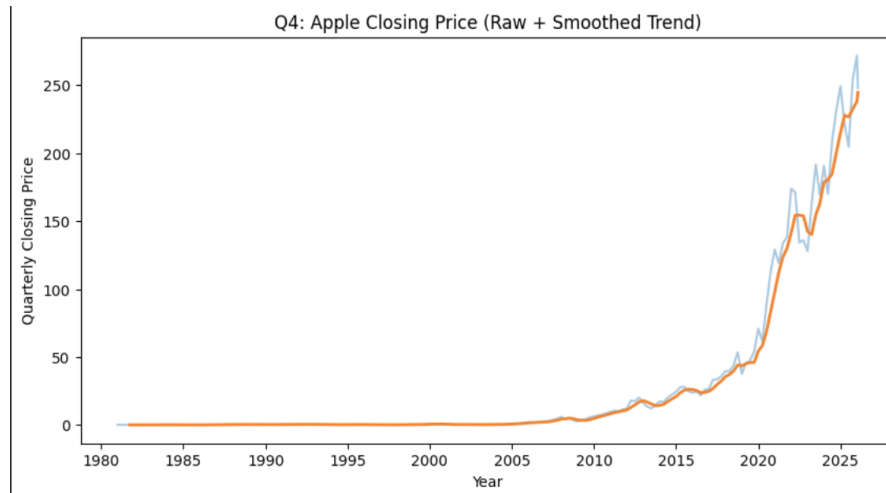
Return volatility is right-skewed, with most values clustering between 0.01 and 0.03, which reflects typical, low-to-moderate daily risk levels for a stable large-cap stock like Apple. A smaller number of periods show volatility rising above 0.04, indicating noticeably elevated uncertainty. The long right tail highlights rare spikes above 0.06, which signal unusually turbulent market conditions.



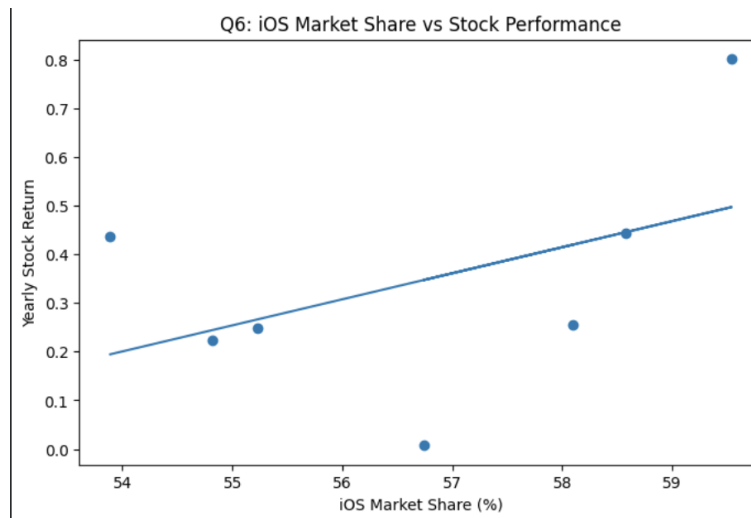
open_price

Apple's opening prices are heavily right-skewed, with most values clustered under \$20 due to historical stock splits. The long tail toward higher prices reflects the company's strong growth over time. Only a few observations appear in the triple-digit range, showing that high opening prices are relatively recent. The distribution highlights Apple's shift from a low-priced stock to a much higher-valued one.

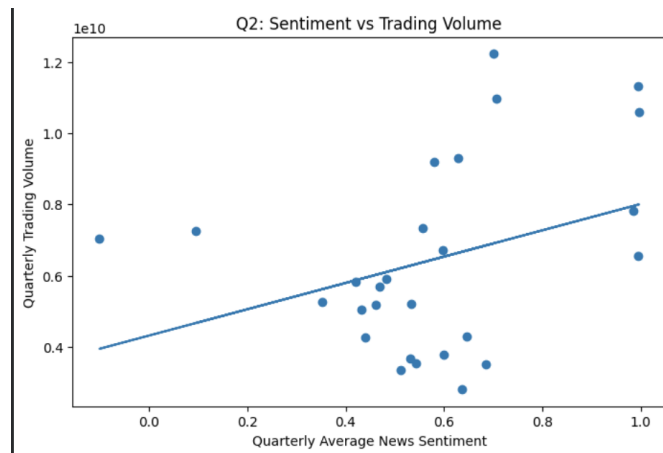
Visual analysis



Multiple time-series charts, especially the long-term stock price trend and iPhone revenue/units sold, show that Apple's financial growth is structurally linked to the iPhone product cycle. The stock price accelerates sharply after the iPhone's introduction, and yearly revenue visuals confirm the iPhone dominated Apple's earnings for more than a decade. These charts highlight Apple's transition from a diversified tech company to one whose valuation is strongly shaped by a single flagship product line.



The iOS vs. Android market share visualizations reveal a surprisingly stable and slightly rising iOS share over time, despite intense global competition. This stability, paired with consistent growth in global iPhone users, suggests Apple's ecosystem creates long-term customer lock-in. Across plots, the steadiness of iOS adoption aligns with Apple's sustained financial performance.



Across the scatterplots and time-series comparisons, sentiment showed a moderate positive relationship with trading volume, meaning quarters with unusually high or low sentiment tended to experience higher trading activity. This pattern suggests that sentiment doesn't directly influence returns, but it does increase investor attention and market participation. When sentiment spikes, whether strongly positive or strongly negative, investors react by trading more, resulting in noticeable volume increases during those periods.

Across scatterplots and normalized time series, a clear pattern emerges: extreme sentiment periods correlate with higher trading volume. Even though sentiment does not predict returns, the charts show that investor activity intensifies during highly positive or negative periods. This indicates sentiment acts as a market attention mechanism, affecting participation rather than price direction.

Stock return and volatility visualizations show that short-term returns behave randomly, with no clear autocorrelation patterns. However, the long-term stock price trend chart reveals smooth, compounding growth. This contrast highlights that while Apple's year-to-year performance is unpredictable, the long-term trajectory is strongly upward, which is consistent with its ecosystem strength, revenue base, and brand perception.

Analytical & statistical methods

To analyze temporal patterns in the Apple datasets, I applied two domain-specific time-series methods from the statsmodels.tsa library: Autocorrelation and Partial Autocorrelation (ACF/PACF) and the Augmented Dickey-Fuller (ADF) Unit Root Test. ACF and PACF were used to evaluate the structure of Apple's yearly stock returns, measuring whether returns depend on prior years or display any persistent autoregressive pattern. The results showed minimal autocorrelation and no meaningful PACF spikes, indicating that Apple's yearly returns behave like a random walk. This suggests that past performance does not reliably predict future outcomes and that Apple's long-run valuation is driven primarily by new market information rather than serial dependence.

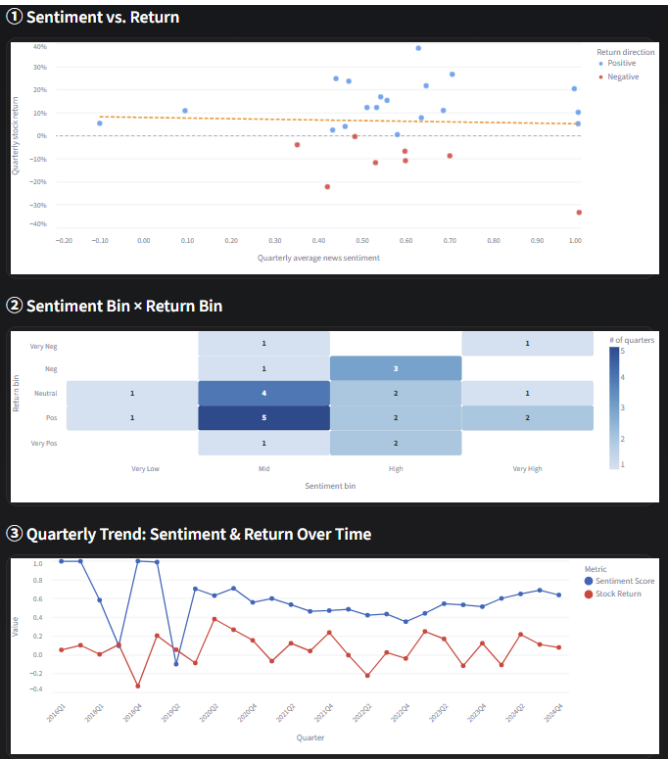
The ADF test was applied to the quarterly average news sentiment series to determine whether sentiment is stationary or trending over time. The test produced a p-value greater than 0.05, meaning the series is non-stationary and exhibits a long-term upward drift rather than fluctuating around a constant mean. This indicates that Apple's media sentiment has gradually strengthened over time, reflecting increasing positive public perception and sustained brand momentum. Together, these two analytical methods provided

statistical confirmation that Apple’s financial performance shows little autoregressive structure, while sentiment shows persistent structural improvement.

Polished visualizations + Enhancements

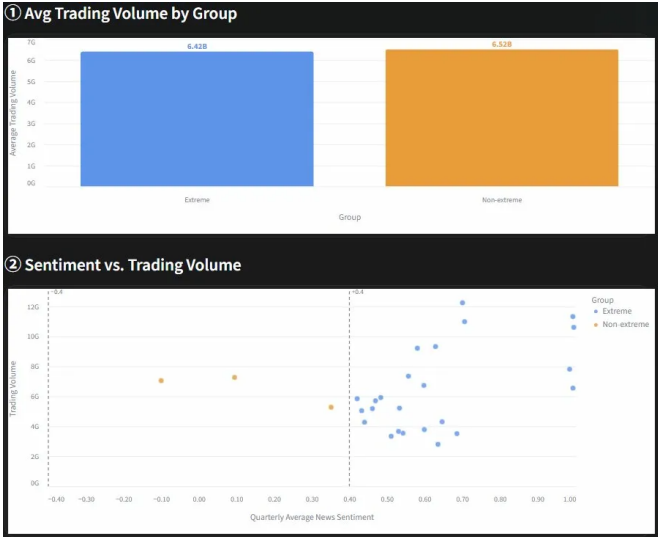
Q1. Is quarterly news sentiment associated with Apple’s quarterly stock return?

To explore the relationship between Apple's news sentiment and stock performance, three complementary visualizations were designed. The scatter plot color-encodes each quarter by return direction — blue for positive, red for negative — with a dashed $y = 0$ baseline and an orange regression line to surface directional patterns at a glance. The heatmap bins both sentiment and return into five quantile groups, using a sequential blue color scheme with overlaid count labels to reveal where quarter observations concentrate across the sentiment-return space. Finally, the dual time-series line chart plots sentiment score and stock return together across quarters, colored by metric, making it easy to compare how the two signals move relative to each other over time. A quarter range slider and regression toggle were moved to the sidebar, and the raw data table was collapsed into an expander to keep the main canvas focused on analysis.



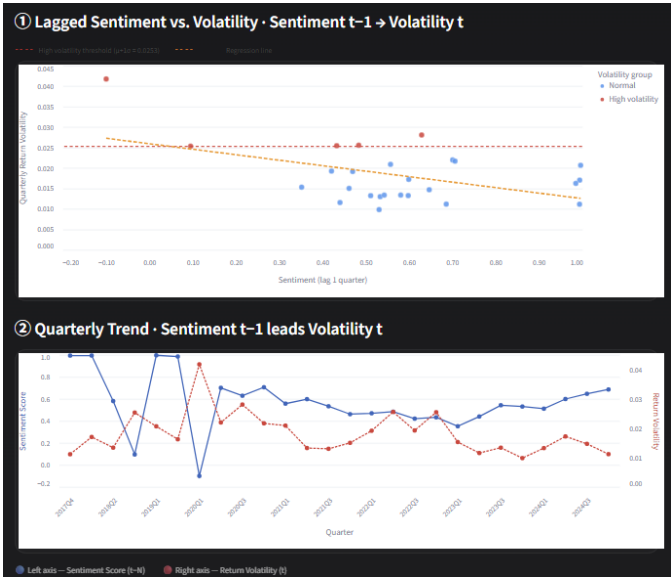
Q2. Do extreme sentiment quarters correspond to higher trading volume?

To examine whether quarters with extreme news sentiment drive higher trading activity, two visualizations were designed. The bar chart compares average trading volume between Extreme and Non-extreme sentiment groups, using distinct colors — blue for Extreme, orange for Non-extreme — with value labels formatted in billion units directly on each bar to make the difference immediately readable. The scatter plot maps each quarter's sentiment score against its trading volume, with points color-coded to match the bar chart's scheme and dashed vertical threshold lines overlaid at $x = \pm \text{threshold}$ to make the active Extreme/Non-extreme boundary explicitly visible on the chart. Summary metrics were replaced with st.metric cards showing extreme quarter count, average volume per group, and the percentage delta between them. The threshold slider and quarter range slider were moved to the sidebar, and the raw data table was collapsed into an st.expander.



Q3. Do sentiment spikes precede larger price volatility?

To investigate whether sentiment leads volatility, two visualizations were designed. The scatter plot color-encodes each quarter by volatility group — blue for Normal, red for High volatility (defined as mean + 1 SD) — with a dashed horizontal threshold line marking that boundary, and an orange dashed regression line styled to contrast clearly against the data layer. An inline reference legend between the chart title and chart body labels both the threshold and regression line directly, so the viewer does not need to cross-reference a separate legend. The dual-axis line chart plots lagged sentiment on the left y-axis and return volatility on the right y-axis, each scaled independently so both series fill their own range and the lead-lag pattern becomes visually traceable across quarters — the chart title updates dynamically to reflect the active lag (e.g. "Sentiment t-1 leads Volatility t"). Summary metrics were replaced with st.metric cards showing quarters analysed, correlation at the active lag, and an interpretation label. The lag slider, quarter range slider, and regression toggle were moved to the sidebar, and the raw data table was collapsed into an st.expander.



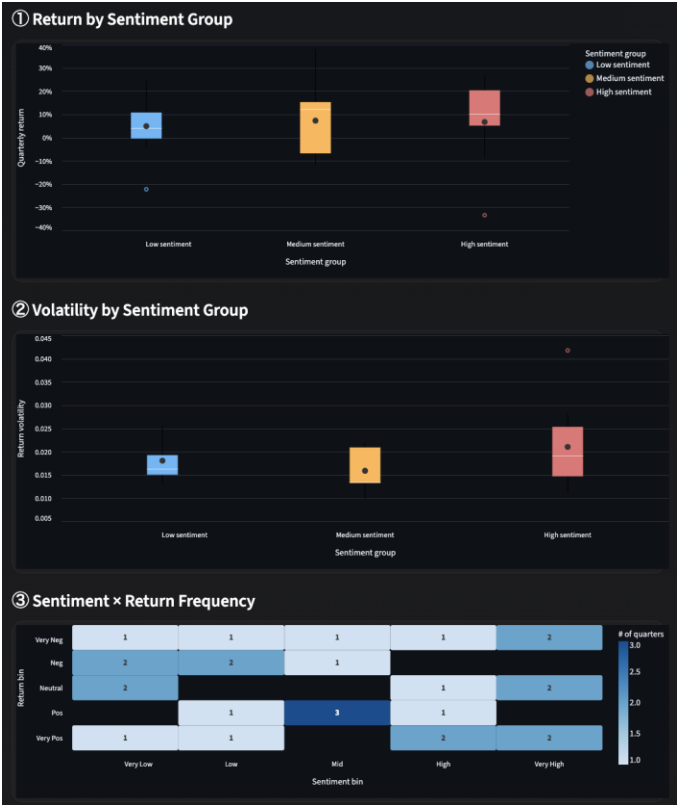
Q4. How has Apple's stock return evolved over time?

For Q4, I enhanced the earlier return visualizations by combining quarterly returns with cumulative returns to better capture both short-term fluctuations and long-term performance. The bar chart highlights variability across individual quarters, clearly showing periods of volatility like the sharp declines in 2018 and 2022, while the cumulative return line smooths out this noise to reveal the overall upward trajectory. By pairing these two views, the chart improves on previous visuals by balancing detail and clarity. The quarterly returns emphasize risk and inconsistency, while the cumulative trend shows that, despite downturns, Apple's stock has experienced strong sustained growth, reaching its highest levels in the most recent periods. The clean design and simple color encoding keep the focus on how returns evolve over time.



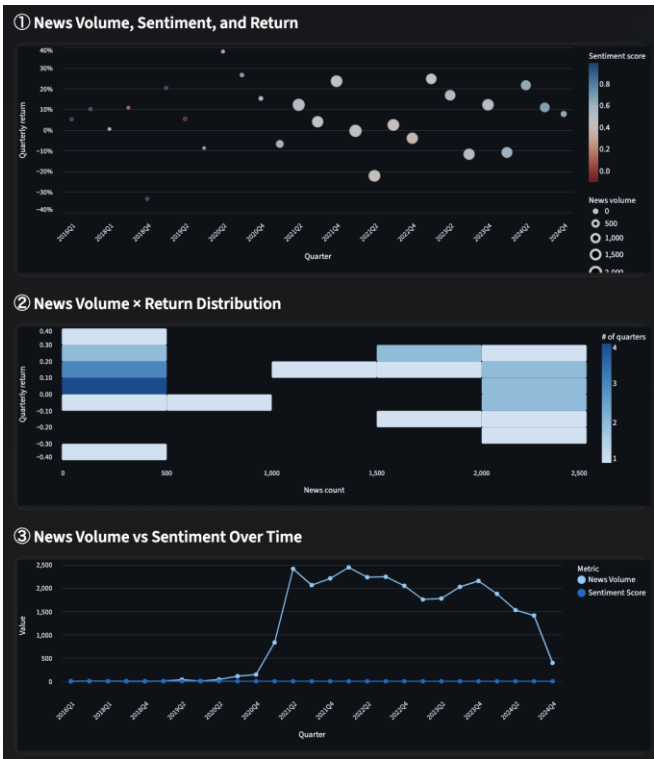
Q5. How do different sentiment levels relate to return and volatility distributions?

I enhanced the earlier sentiment analysis by combining distribution-focused boxplots with a frequency heatmap to capture both summary patterns and detailed relationships. The return and volatility boxplots show how performance varies across low, medium, and high sentiment groups, making it easy to compare central tendencies and spread. For example, higher sentiment is generally associated with higher returns, but also with greater variability, especially in volatility. To complement this, the heatmap adds a more granular view by showing how often different sentiment levels align with specific return ranges. This helps reveal that while higher sentiment is more frequently linked with positive returns, there is still overlap across categories, indicating the relationship is not perfectly linear. By pairing these views, the visualization improves on earlier charts by balancing clarity and depth. The boxplots summarize distributional differences, while the heatmap uncovers underlying frequency patterns, making the relationship between sentiment, return, and volatility more interpretable and aligned with effective visual storytelling.



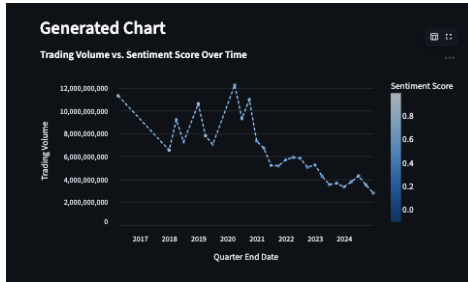
Q6. How does news volume relate to sentiment and stock performance over time?

For this question, I enhanced the analysis by combining a multi-encoding scatterplot with supporting distribution and time-series views to show how news volume interacts with sentiment and returns. The first chart uses size (news volume) and color (sentiment) to highlight that higher news volume often coincides with more extreme outcomes - both positive and negative - rather than consistently higher returns. The heatmap further shows that higher news counts are associated with a wider spread of returns, reinforcing increased volatility. The time-series view adds context by showing that spikes in news volume don't always align with stronger sentiment, but often occur during periods of market activity or uncertainty. Together, these views show that news volume is more closely tied to variability in performance, while sentiment plays a more nuanced role within those high-activity periods.

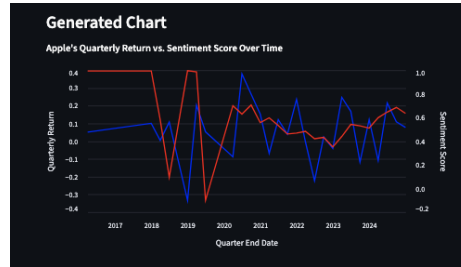


LLM-generated visualizations

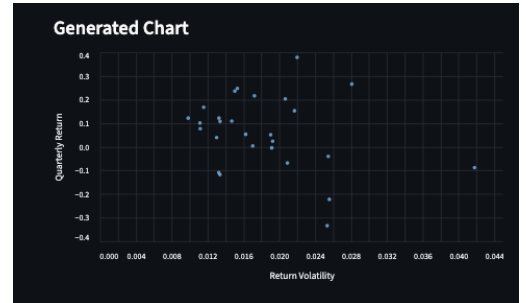
1) 'Does extreme sentiment correspond to higher trading volume?'



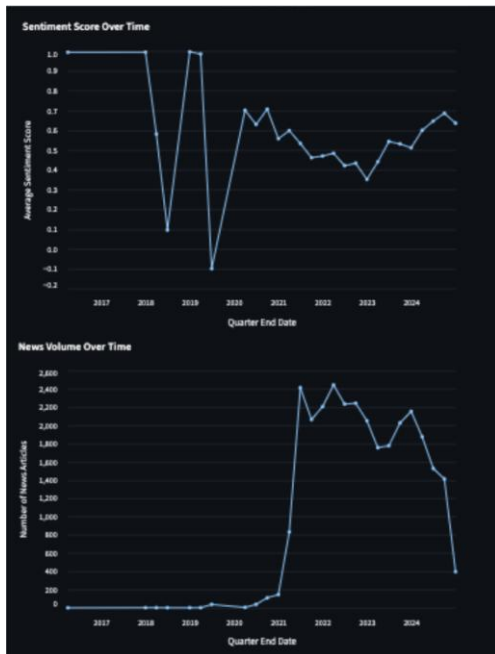
2) 'What is Apple's trend in stock compared to sentiment over time?'



3) 'Show the association between volatility and returns'



4) 'Does news volume relate to Apple's overall financial success?'



Deep Learning Inference Analysis

Milestone 8 applied three zero-shot foundation models to forecast Apple's quarterly closing stock price, extending the earlier sentiment and price analysis into forward-looking inference.

Key findings:

- **Gemini 1.5 Flash** (price + sentiment, MAE \$18.28) outperformed price-only models, confirming that news sentiment carries measurable predictive signal.
- **Chronos-2 Bolt** (MAE \$28.57) generalized consistently across 2024 and 2025 despite using price history only.
- **TimesFM 2.0** suffered scale distortion from Apple's long-term price growth (~\$0.10 → ~\$250), producing inflated forecasts — a key limitation on low-frequency, high-growth series.
- All three models missed the Q4 2024 and Q2 2025 price drops, confirming that external market shocks remain difficult to capture zero-shot.

Sample Visualizations

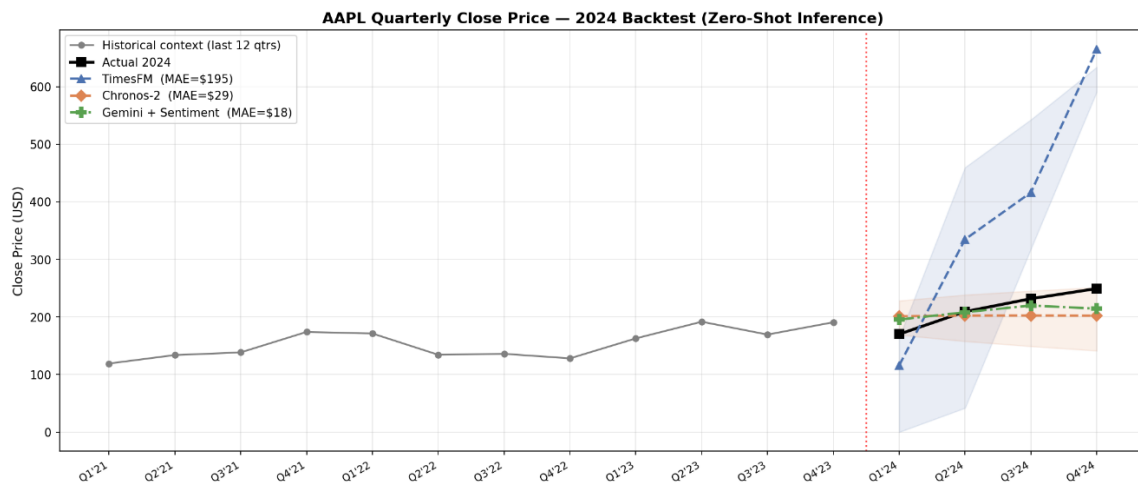


Figure 1. 2024 backtest. Chronos-2 and Gemini track actual prices closely; TimesFM overextrapolates due to full historical context (1980–2023).

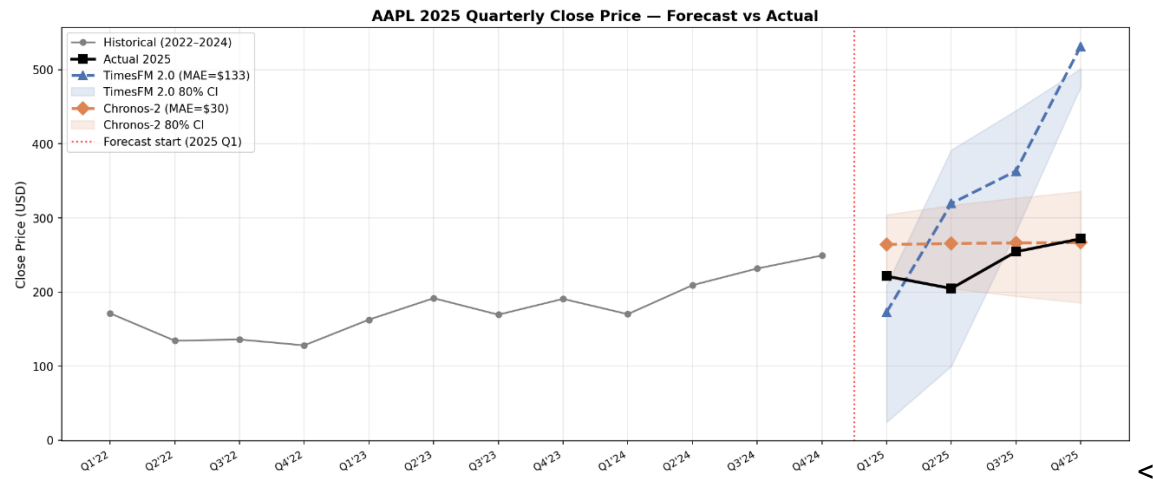


Figure 2. 2025 forecast vs. actual. Chronos-2 predicts a flat trajectory near \$265 (MAE \$30); TimesFM overestimates throughout (MAE \$133).

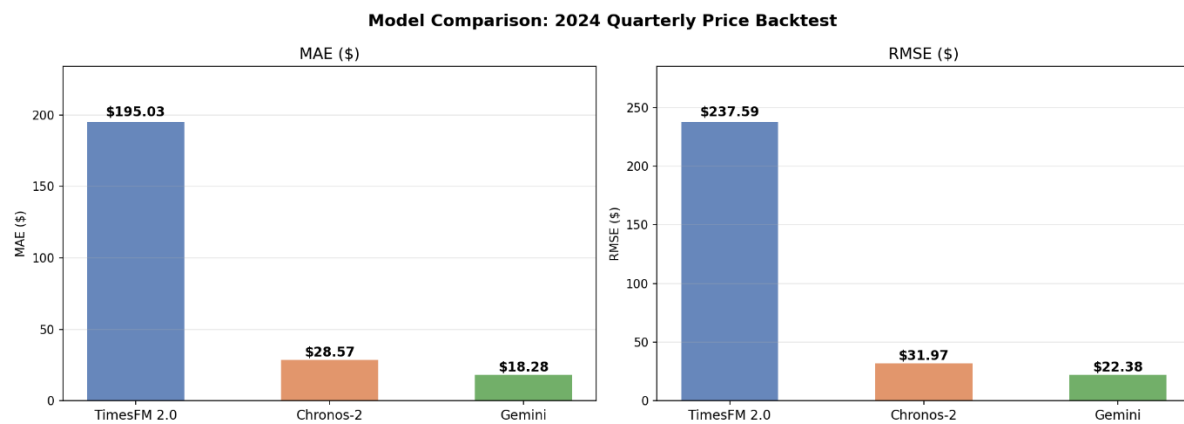


Figure 3. Model comparison on 2024 backtest: Gemini (MAE \$18) > Chronos-2 (MAE \$29) > TimesFM (MAE \$195).

Cleaning and processing

Missing values

To identify missing values we used `.isnull()` and `.sum()` to determine the extent and location of incomplete data. In numerical columns, missing values were handled using certain strategies depending on the data context. For time-series variables, interpolation or forward/backward filling was applied to maintain trends and continuity. For short gaps we used forward fill due to our assumption of short-term stability. In situations where missing observations were minimal or not critical, the rows were removed using `.dropna()`. We used these methods to ensure that all data points were filled and consistent for future analysis.

Aggregation

We used aggregation to summarize the dataset at meaningful levels for analysis. Using `groupby()` and `resample()` functions, data was aggregated across times (monthly/yearly totals) and across categorical variables (state or region). Common aggregation metrics included sums, means, and totals. `sum()` was used to aggregate sales-related data to reflect total monthly performance, while `mean()` was used to reflect averages of continuous environmental variables. This step helped reduce data granularity, which allowed for clear trend analysis/comparisons across groups while maintaining the integrity of KPIs.

Joins of input sources

The final step of preparing our data set included merging and joining operations in pandas across multiple datasets. The datasets were combined based on shared keys such as date or categorical identifiers. Appropriate join types like inner, left, or outer joins were selected depending on whether only matching records or all observations were required. After merging, the resulting dataset was validated to check for introduced missing values or inconsistencies. This integration helped further our analysis by combining complementary features from multiple data sources into a single analytical framework.